Databases: Adding, Changing and Deleting Data

Create tables and keep your data up to date, safely.

Databases | Beginner to Intermediate | Ages 12–18

💡 The Big Idea

- SELECT reads data. These commands change it.
- `INSERT` adds new rows, `UPDATE` edits existing rows and `DELETE` removes rows.
- `CREATE TABLE` makes a new table.
- Always use `WHERE` with `UPDATE` and `DELETE`, or every row changes.

➕ Add and Create

- `CREATE TABLE name (column type, ...);` makes a table.
- Common types: `INTEGER` for whole numbers, `TEXT` for words.
- `INSERT INTO table (cols) VALUES (...);` adds a row.
- A `PRIMARY KEY` is a unique id for each row.

✏️ Edit and Remove

- `UPDATE table SET col = value WHERE ...;` changes rows.
- `DELETE FROM table WHERE ...;` removes rows.
- Use `SELECT` with the same `WHERE` first to preview which rows will change.
- Back up important data before big changes.

💻 Code Examples

Create a table SQL

```sql
CREATE TABLE pets (id INTEGER PRIMARY KEY, name TEXT, age INTEGER);
INSERT INTO pets (name, age) VALUES ('Coco', 3);
SELECT * FROM pets;
```

▶ **Output:**

```
OK, done.

(1 row changed)

id  name  age
--  ----  ---
1   Coco  3
```

Creates a new pets table, adds Coco, and reads it back. The id fills itself in.

Add a student SQL

```sql
INSERT INTO students (name, age, class, score) VALUES ('Isha', 13, '7B', 78);
SELECT * FROM students;
```

▶ **Output:**

```
(1 row changed)

id  name   age  class  score
--  -----  ---  -----  -----
1   Aarav  12   6A     88
2   Mia    13   7B     95
3   Zoya   12   6A     72
4   Kabir  14   8C     64
5   Anaya  13   7B     81
6   Dev    12   6B     90
7   Isha   13   7B     78
```

The id is filled in automatically because it is the primary key.

Update one student SQL

```sql
UPDATE students SET score = 85 WHERE name = 'Zoya';
SELECT name, score FROM students WHERE name = 'Zoya';
```

▶ **Output:**

```
(1 row changed)

name  score
----  -----
Zoya  85
```

Zoya's score changes from 72 to 85.

Preview, then delete SQL

```sql
SELECT * FROM students WHERE id = 4;
DELETE FROM students WHERE id = 4;
SELECT COUNT(*) AS total FROM students;
```

▶ **Output:**

```
id  name   age  class  score
--  -----  ---  -----  -----
4   Kabir  14   8C     64

(1 row changed)

total
-----
5
```

Kabir is removed, so 5 students remain.

Danger: no WHERE SQL

```sql
UPDATE students SET score = 0;
SELECT name, score FROM students;
```

▶ **Output:**

```
(6 rows changed)

name   score
-----  -----
Aarav  0
Mia    0
Zoya   0
Kabir  0
Anaya  0
Dev    0
```

Without WHERE, every single row changes. Always check your WHERE.

📚 Key Terms

INSERT
Adds a new row.

CREATE TABLE
Makes a new table.

UPDATE
Changes existing rows.

Primary key
A unique id for each row.

DELETE
Removes rows.

Data type
The kind of data a column holds.

INTEGER
A whole number.

TEXT
Words and letters.

WHERE
Chooses which rows are affected.

Backup
A saved copy of your data.

🧮 Quick Reference

```
CREATE TABLE t (id INTEGER PRIMARY KEY, name TEXT);
INSERT INTO t (name) VALUES ('Mia');
UPDATE t SET name = 'Zoe' WHERE id = 1;
DELETE FROM t WHERE id = 1;
-- Always preview with SELECT first
```

✏️ Worked Examples

Q: Add a student called Isha, age 13, class 7B, score 78.

A: INSERT INTO students (name, age, class, score) VALUES ('Isha', 13, '7B', 78);

Q: Change Dev's score to 92.

A: UPDATE students SET score = 92 WHERE name = 'Dev'; The WHERE clause keeps other students unchanged.

Q: Remove the student with id 3.

A: DELETE FROM students WHERE id = 3; Preview first with SELECT * FROM students WHERE id = 3;

🚫 Common Mistakes

- Running `UPDATE` or `DELETE` with no `WHERE`.
- Putting text values without quotes.
- Giving two rows the same primary key.
- Mixing up the order of the columns and the values in `INSERT`.
- Forgetting that deleted data may be gone for good.

👨‍👩‍👧 Parent Corner

- Use a sandbox or practice database so nothing important can be lost.

- Ask your child: What could go wrong if I forget WHERE?
- Compare DELETE to using an eraser: it can't be undone in many systems.
- Praise the habit of previewing with SELECT first.
- Real websites, apps and games all use these four actions: create, read, update, delete.

🎯 Quick Check

1. Which command adds a new row?

✅ INSERT.

2. Which command changes existing data?

✅ UPDATE.

3. What happens if UPDATE has no WHERE?

✅ Every row changes.

4. What is a primary key?

✅ A unique id for each row.

5. How can you preview a delete safely?

✅ Run a SELECT with the same WHERE first.